PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	**Performance Testing**	DATE	July 2026

Performance Testing

Model Evaluation Metrics

All four models were evaluated on a held-out 20% test set (7,292 applicants) using precision, recall, F1-score, and overall accuracy, with particular focus on the minority "Rejected" class given the ~88/12 class imbalance.

Model	Precision (Rejected)	Recall (Rejected)	F1-score (Rejected)	Accuracy
Logistic Regression	0.13	0.50	0.21	0.56
Decision Tree	0.31	0.66	0.42	0.79
Random Forest (Selected)	0.37	0.54	0.44	0.84
XGBoost	0.27	0.58	0.37	0.76

Selected Model

Random Forest was selected as the final model based on its best F1-score for the rejected class (0.44) and highest overall accuracy (0.84), representing the best balance between correctly identifying risky applicants and minimizing false rejections of low-risk applicants.

Application Testing

- Verified the Flask application starts correctly and serves all three pages
- Verified form submission correctly maps to the model's expected feature order
- Tested with a known test-set applicant record correctly classified as "Rejected" by the model, and confirmed the web application returned the same result end-to-end
- Confirmed the application also returns "Approved" results for low-risk profiles